



Dry lab notebook

Hydrophobin design

I. Preparation

[1/08/2023 - 21/08/2023]

Purpose of the protein design: we want to design our own hydrophobin with a lower isoelectric point in order to emulsify more efficiently in lower pH values. Hydrophobins emulsify the best in alkaline pH (around 9) whereas *Yarrowia lipolytica* prefers lower pH values. In addition, unless the fermentation is done in a bioreactor with a constant pH, the pH drops very rapidly.

Information regarding hydrophobins was provided by Jeroen Vereman and the information on *Yarrowia lipolytica* from Vasileios Vangalis and Seppe Dockx from prof. Kevin Verstrepen's lab.

Strategy for the design of a hydrophobin with a lower pI was proposed by our PI, Vitor Pinheiro.

II. Generating sequences with lower theoretical pI

[Script for substitution of residues](#)

[22/08/2023 - 24/08/2023]

As a starting point, we wrote a python script that generates all possible combinations of the original sequence where lysine and arginine is substituted with either glutamate or aspartate.

The script uses BioPython¹ to calculate the pI of the protein.

Initial sequence: HFBI – 75 amino acids (2 lysines and 1 arginine)

PDB code: 2FZ6

Result: 26 modified sequences, pI ranging from 4.05 - 4.22 with default interval of pH to investigate, when the minimum of the interval is lowered, the pI goes down even to 3.17 (lowest pI)

24/08/2023:

Generated sequences with 3 (or less) random leucines substituted with aspartates and a sequence with all 7 leucines substituted with aspartate.

III. Investigating the impact of the mutations on the structure

Prediction of the structure of modified sequences with AlphaFold

[24/08/2023 – 31/08/2023]

Predicted structures of:

- Sequence with lowest pI after mutating (lysine and arginine are substituted with either glutamate or aspartate) the original HFBI sequence
- Original sequence with 3 random leucines substituted with aspartate (mutations: L24D, L67D, L68D)
- Sequence with 3 leucines substituted with aspartate (L24D, L67D, L68D) further mutated to have the lowest pI
- Original sequence with all 7 leucines substituted with aspartate

Figure below shows superposition of the original HFBI and HFBI with 3 random leucines substituted with aspartate. We expected the mutations to highly impact the stability of the protein and result in either a very different structure or unstructured regions but we ended up with a very similar structure with some (but not very substantial) differences (e.g., in the β -sheets region). This points to one of the limitations of AlphaFold². In particular, it does not predict the impact of mutations on the protein structure or the stability of the protein.

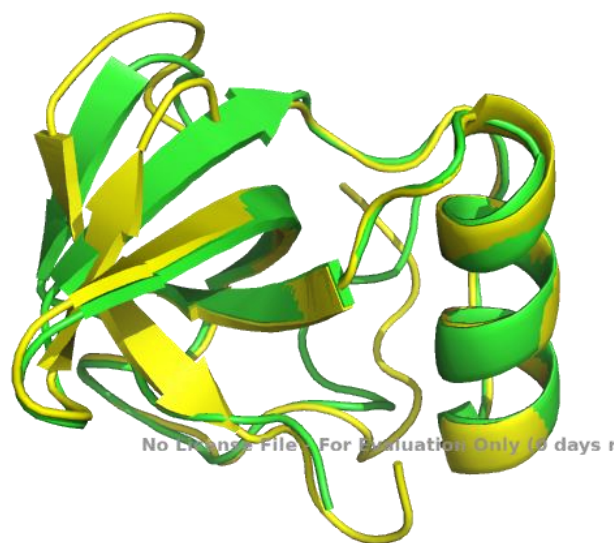


Figure 1. Superposition of HFBI (green) and HFBI with three random leucines substituted with aspartate (yellow).

In order to fully explore and confirm the encountered mutations, we predicted a structure of the HFBI with all seven leucines substituted with aspartate. Figure below shows this structure aligned to the original HFBI structure. Obtained structure is still very similar to the original one and does not display any vast differences. This result further confirms that AlphaFold is not a good tool (at least yet) to study the impact of mutations on the protein stability and structure.

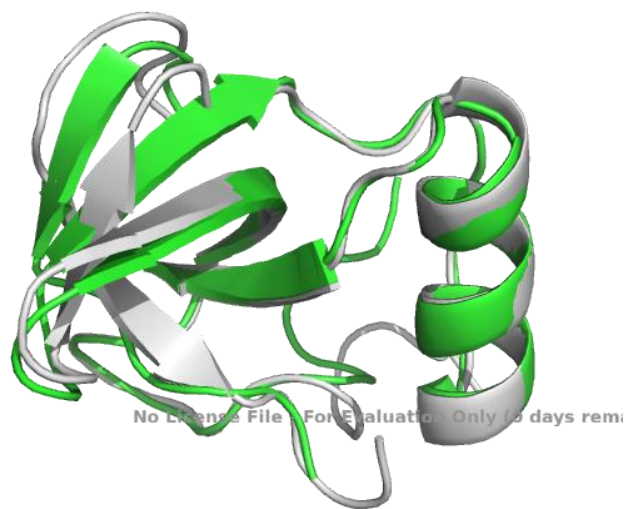


Figure 2. Superposition of HFBI (green) and HFBI with all seven leucines substituted with aspartate (grey).

As a result of these conclusions, we decided to no longer use AlphaFold for designing a hydrophobin with lower isoelectric point and we are looking for other strategies to tackle this idea.

Prediction of disease-related mutations with PredictSNP

[04/09/2023 - 08/09/2023]

PredictSNP³ is a tool that predicts whether a given mutation is disease-related, indicating its potential impact on the stability of the protein. We submitted four jobs to PredictSNP, each involving a modification of the original HFBI sequence, all of which corresponded to the four sequences previously submitted to AlphaFold.

Note: The algorithm is trained on human protein data, while hydrophobins are fungal proteins. Therefore, we can only assume its accuracy for hydrophobins, but it is not necessarily guaranteed.

Mutated HFBI with lowest pI

Most of the tools integrated into PredictSNP suggest that the mutations are likely to be neutral. The expected accuracy of PredictSNP is approximately 75% for each mutation. The MAPP prediction raises some concern as its confidence level is approximately 78%. Further exploration of the sequence and the impact of these mutations is necessary. These initial results are promising because there is a possibility that these mutations may not significantly affect the protein's stability, which aligns with our research objectives.

PredictSNP result:

Mutation	PredictSNP prediction	MAPP prediction	PhD-SNP prediction	PolyPhen-1 prediction	PolyPhen-2 prediction	SIFT prediction	SNAP prediction
K32D	NEUTRAL	DELETERIOUS	NEUTRAL	NEUTRAL	NEUTRAL	NEUTRAL	NEUTRAL
R45D	NEUTRAL	DELETERIOUS	NEUTRAL	NEUTRAL	DELETERIOUS	NEUTRAL	DELETERIOUS
K50D	NEUTRAL	DELETERIOUS	NEUTRAL	NEUTRAL	NEUTRAL	NEUTRAL	NEUTRAL

Original sequence with 3 random leucines substituted with aspartate

All three mutations appear to be deleterious, indicating that they indeed have an impact on the stability of the protein. This result aligns with our expectations, as these mutations were intentionally designed to affect the protein's structure.

PredictSNP result:

Mutation	PredictSNP prediction	MAPP prediction	PhD-SNP prediction	PolyPhen-1 prediction	PolyPhen-2 prediction	SIFT prediction	SNAP prediction
L24D	DELETERIOUS	DELETERIOUS	DELETERIOUS	DELETERIOUS	DELETERIOUS	DELETERIOUS	DELETERIOUS
L67D	DELETERIOUS	DELETERIOUS	NEUTRAL	DELETERIOUS	DELETERIOUS	DELETERIOUS	DELETERIOUS
L68D	DELETERIOUS	DELETERIOUS	NEUTRAL	DELETERIOUS	DELETERIOUS	DELETERIOUS	DELETERIOUS

Sequence with 3 leucines substituted further mutated towards lower pI

This sequence contains mutations that reduce the pI and involve the substitution of leucines with aspartate, both of which are expected to disrupt protein stability. This result consolidates the findings from the previous two sequences.

PredictSNP result:

Mutation	PredictSNP prediction	MAPP prediction	PhD-SNP prediction	PolyPhen-1 prediction	PolyPhen-2 prediction	SIFT prediction	SNAP prediction
L24D	DELETERIOUS	DELETERIOUS	DELETERIOUS	DELETERIOUS	DELETERIOUS	DELETERIOUS	DELETERIOUS
K32D	NEUTRAL	DELETERIOUS	NEUTRAL	NEUTRAL	NEUTRAL	NEUTRAL	NEUTRAL
R45D	NEUTRAL	DELETERIOUS	NEUTRAL	NEUTRAL	DELETERIOUS	NEUTRAL	DELETERIOUS

K50D	NEUTRAL	DELETERIOUS	NEUTRAL	NEUTRAL	NEUTRAL	NEUTRAL	NEUTRAL
L67D	DELETERIOUS	DELETERIOUS	NEUTRAL	DELETERIOUS	DELETERIOUS	DELETERIOUS	DELETERIOUS
L68D	DELETERIOUS	DELETERIOUS	NEUTRAL	DELETERIOUS	DELETERIOUS	DELETERIOUS	DELETERIOUS

All leucines substituted

In this sequence, all leucines have been replaced with aspartate. We made this choice to challenge the limitations of AlphaFold. As anticipated, these mutations are expected to influence the protein's structure, a prediction supported by PredictSNP but not detected by AlphaFold. Upon comparing the prediction confidence from all the tools, it is reasonable to assume that all of these mutations would indeed affect the protein's stability.

PredictSNP result:

Mutation	PredictSNP prediction	MAPP prediction	PhD-SNP prediction	PolyPhen-1 prediction	PolyPhen-2 prediction	SIFT prediction	SNAP prediction
L12D	DELETERIOUS	DELETERIOUS	DELETERIOUS	DELETERIOUS	DELETERIOUS	DELETERIOUS	DELETERIOUS
L24D	DELETERIOUS	DELETERIOUS	DELETERIOUS	DELETERIOUS	DELETERIOUS	DELETERIOUS	DELETERIOUS
L26D	DELETERIOUS	DELETERIOUS	NEUTRAL	DELETERIOUS	DELETERIOUS	DELETERIOUS	DELETERIOUS
L29D	DELETERIOUS	DELETERIOUS	DELETERIOUS	DELETERIOUS	DELETERIOUS	DELETERIOUS	DELETERIOUS
L56D	NEUTRAL	DELETERIOUS	NEUTRAL	DELETERIOUS	DELETERIOUS	NEUTRAL	NEUTRAL
L67D	DELETERIOUS	DELETERIOUS	NEUTRAL	DELETERIOUS	DELETERIOUS	DELETERIOUS	DELETERIOUS
L68D	DELETERIOUS	DELETERIOUS	NEUTRAL	DELETERIOUS	DELETERIOUS	DELETERIOUS	DELETERIOUS

Next steps

The analysis we have conducted so far is preliminary. Further analysis and exploration are necessary to gain greater confidence in our designed hydrophobin. Our next steps would involve utilizing FoldX^{4,5} to thoroughly assess the stability of the mutated HFBI. By applying a force field to our hydrophobin, we could investigate the impact of mutations on both the stability and dynamics of the protein.

In the earlier stages, we addressed one of the limitations of AlphaFold, which is its inability to predict the effect of single nucleotide polymorphisms on protein structure. In September 2023, Google DeepMind, the creators of AlphaFold, introduced AlphaMissense⁶. This tool, built upon AlphaFold, is designed to predict whether a given mutation is disease-related, potentially disrupting protein stability. Regrettably, we haven't had the opportunity to test it ourselves, but it presents an exciting avenue for our future analysis.



References

1. Cock, P. J. A. *et al.* Biopython: Freely available Python tools for computational molecular biology and bioinformatics. *Bioinformatics* **25**, (2009).
2. Jumper, J. *et al.* Highly accurate protein structure prediction with AlphaFold. *Nature* **596**, (2021).
3. Bendl, J. *et al.* PredictSNP: Robust and Accurate Consensus Classifier for Prediction of Disease-Related Mutations. *PLoS Comput Biol* **10**, (2014).
4. Schymkowitz, J. *et al.* The FoldX web server: An online force field. *Nucleic Acids Res* **33**, (2005).
5. Radusky, L. G. & Serrano, L. PyFoldX: enabling biomolecular analysis and engineering along structural ensembles. *Bioinformatics* **38**, (2022).
6. Jun Cheng et al., Accurate proteome-wide missense variant effect prediction with AlphaMissense. *Science* **381**, eadg7492 (2023). DOI:10.1126/science.adg7492